GABRIELLA'S MATH LAB · DECIMALS WORKBOOK

SECTION 4

Adding Decimals

Line up the decimal points, add each place, and regroup when a place gets to 10.

SKILL
One new idea at a time.

SHOW HOW
Watch a problem solved step by step.

TRY ONE
Finish a problem with help.

YOUR TURN
Solve problems on your own.

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Name _____ Started _____

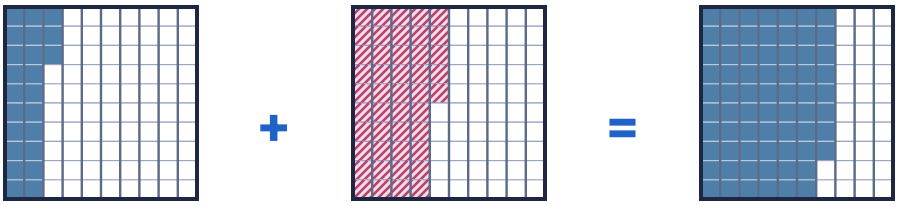
Georgia K-12 Mathematics Standards 5.NR.4.4 (add decimals to the hundredths place using a variety of strategies).

4.1 Adding with Place Value

Goal: I can add tenths to tenths and hundredths to hundredths.

SHOW HOW

Add 0.23 + 0.45.



0.23 + 0.45 = 0.68

1. **Tenths with tenths:** 2 tenths + 4 tenths = **6 tenths** (the full columns).
2. **Hundredths with hundredths:** 3 hundredths + 5 hundredths = **8 hundredths** (the extra squares).
3. Put them together: 6 tenths and 8 hundredths = **0.68**

Why it works: you can only add things of the same size, like adding apples to apples. Tenths add to tenths; hundredths add to hundredths.

LET'S TRY ONE

Add 0.52 + 0.31.

Tenths: 5 + 3 = _____ tenths Hundredths: 2 + 1 = _____ hundredths

Sum: _____

YOUR TURN — SKILL 4.1 · ADD PLACE BY PLACE

1 **0.2 + 0.5**

Sum _____

2 **0.6 + 0.3**

Sum _____

3 **0.14 + 0.23**

Sum _____

4 **0.41 + 0.36**

Sum _____

5 **1.2 + 3.4** (ones with ones, too)

Sum _____

6 **2.13 + 4.05**

Sum _____

7 **0.305 + 0.142**

Sum _____

8 **3.21 + 1.6** (1.6 has 0 hundredths)

Sum _____

4.2 Line Up and Add

Goal: I can line up the decimal points and add in columns.

SHOW HOW

Add 3.42 + 1.35.

```
  3.42
+ 1.35
------
  4.77
```

1. **Line up the decimal points**, one under the other. Then every place is in its own column.
2. Start at the **right**. Hundredths: 2 + 5 = **7**.
3. Tenths: 4 + 3 = **7**. Ones: 3 + 1 = **4**.
4. Bring the **decimal point straight down**: **4.77**

Why it works: lining up the decimal points lines up the places, so you add hundredths to hundredths and tenths to tenths, just like in 4.1.

LET'S TRY ONE

Add 5.21 + 2.36. Fill in the boxes.

```
  5.21
+ 2.36
------
   .
```

Hundredths: _____ Tenths: _____ Ones: _____

Sum: _____

YOUR TURN

SKILL 4.2 · ADD. FILL IN THE BOXES.

1

$$\begin{array}{r} 2.31 \\ +\ 1.45 \\ \hline \square\square.\square\square \end{array}$$

2

$$\begin{array}{r} 4.12 \\ +\ 3.76 \\ \hline \square\square.\square\square \end{array}$$

3

$$\begin{array}{r} 0.53 \\ +\ 0.24 \\ \hline \square\square.\square\square \end{array}$$

4

$$\begin{array}{r} 7.06 \\ +\ 2.81 \\ \hline \square\square.\square\square \end{array}$$

5

$$\begin{array}{r} 12.3 \\ +\ \ 5.4 \\ \hline \square\square.\square \end{array}$$

6

$$\begin{array}{r} 6.205 \\ +\ 1.432 \\ \hline \square\square.\square\square\square \end{array}$$

7 Write it in columns, then add: **15.42 + 3.15**

Sum _____

8 Write it in columns, then add: **0.614 + 0.235**

Sum _____

4.3 Adding with Regrouping

Goal: I can regroup 10 of one place as 1 of the next place.

SHOW HOW

Solve 3.45 + 2.27.

```
   1
  3.45
+ 2.27
------
  5.72
```

1. **Line up the decimal points.**
2. **Add the hundredths**: 5 + 7 = 12 hundredths. That is 1 tenth and 2 hundredths: write **2**, carry the *1* to the tenths.
3. **Add the tenths**, including the carried 1: 4 + 2 + 1 = **7**.
4. **Add the ones**: 3 + 2 = **5**.
5. **Place the decimal point** straight down: **5.72**

Why it works: 10 hundredths make 1 tenth (Section 1). So 12 hundredths is 1 tenth + 2 hundredths, and the extra tenth moves to the tenths column.

LET'S TRY ONE

Solve 4.68 + 2.15. Write the carried digit in the small space above.

```
  4.68
+ 2.15
------
  [ ].[ ][ ]
```

Hundredths: 8 + 5 = _____ → write _____, carry _____

Tenths: 6 + 1 + _____ = _____ Sum: _____

YOUR TURN — SKILL 4.3 · ADD. REGROUP WHEN YOU NEED TO.

1

$$\begin{array}{r} 2.58 \\ +\ 1.34 \\ \hline \square\square.\square\square \end{array}$$

2

$$\begin{array}{r} 0.48 \\ +\ 0.39 \\ \hline \square\square.\square\square \end{array}$$

3

$$\begin{array}{r} 5.67 \\ +\ 2.85 \\ \hline \square\square.\square\square \end{array}$$

4

$$\begin{array}{r} 9.6 \\ +\ 2.7 \\ \hline \square\square\square.\square \end{array}$$

5 Write it in columns, then add: **6.75 + 3.48**

Sum _____

6 Write it in columns, then add: **4.98 + 3.06**

Sum _____

7 Write it in columns, then add: **12.85 + 7.36**

Sum _____

8 Write it in columns, then add: **0.587 + 0.256**

Sum _____

4.4 Different Numbers of Decimal Places

Goal: I can add decimals like 4.6 + 2.35 by writing placeholder zeros.

SHOW HOW

Add 4.6 + 2.35.

$$\begin{array}{r} 4.60 \\ +\ 2.35 \\ \hline 6.95 \end{array}$$

1. **Line up the decimal points**, not the last digits. The 6 goes over the 3 (both tenths).
2. 4.6 has no hundredths digit. Write a **placeholder zero**: 4.6 = 4.60 (Section 1: equivalent decimals).
3. Add: hundredths 0 + 5 = 5, tenths 6 + 3 = 9, ones 4 + 2 = 6 → **6.95**

Watch out: lining up the right edges (4.6 over 2.35 with the 6 over the 5) adds tenths to hundredths, which gives a wrong answer. A whole number has its point at the end: 5 = 5.00.

LET'S TRY ONE

Add 3.5 + 1.27. Write the placeholder zero first.

$$\begin{array}{r} 3.5 \\ +\ 1.27 \\ \hline \square.\square\square \end{array}$$

3.5 = _____

Sum: _____

YOUR TURN — SKILL 4.4 · LINE UP THE POINTS, WRITE ZEROS, ADD

1 2.4 + 1.35

Sum _____

2 0.6 + 0.47

Sum _____

3 5 + 2.38

Sum _____

4 3.25 + 4.8

Sum _____

5 0.08 + 0.9

Sum _____

6 9 + 0.75

Sum _____

7 7.125 + 1.6

Sum _____

8 12.4 + 3.756

Sum _____

4.5 Estimate to Check

Goal: I can round to estimate a sum and check that my answer makes sense.

SHOW HOW

Is 5.72 a reasonable answer for 3.45 + 2.27?

1. **Round each number** to the nearest whole number (Section 3): 3.45 ≈ **3** and 2.27 ≈ **2**.
2. **Add the rounded numbers:** 3 + 2 = **5**. The exact sum should be close to 5.
3. **Compare:** 5.72 is close to 5, so it is **reasonable**. If you had written 57.2 or 0.572, the estimate shows the decimal point is in the wrong place.

Why it works: rounded numbers are easy to add in your head, and they are close to the real numbers, so their sum is close to the real sum. An estimate catches decimal-point mistakes.

LET'S TRY ONE

Estimate 6.8 + 2.15 by rounding to whole numbers.

6.8 ≈ _____ 2.15 ≈ _____ Estimate: _____ + _____ = _____

Bonus: the exact sum is 8.95. Is it close to your estimate? _____

YOUR TURN — SKILL 4.5 · ESTIMATE BY ROUNDING TO WHOLE NUMBERS

1 **4.8 + 3.1**

Estimate _____

2 **2.3 + 5.6**

Estimate _____

3 **7.45 + 1.92**

Estimate _____

4 **12.6 + 3.4**

Estimate _____

5 **0.9 + 8.2**

Estimate _____

6 **19.75 + 4.5**

Estimate _____

7 Ben says **3.85 + 3.2 = 70.5**. Estimate to decide if his answer is reasonable.

Reasonable? _____

8 Ana says **5.86 + 2.25 = 8.11**. Estimate to decide if her answer is reasonable.

Reasonable? _____

4.6 Addition Word Problems

Goal: I can tell when a story needs adding, then add and label the answer.

SHOW HOW

Leo ran 2.6 km on Monday and 3.75 km on Tuesday. How far did he run in all?

1. **What is asked?** The total distance. “In all” and “total” mean **add**.
2. **Estimate:** 3 + 4 = about **7 km**.
3. **Line up and add:** 2.60 + 3.75 = **6.35**.
4. **Check and label:** 6.35 is close to 7. Leo ran **6.35 km**.

```
  1
  2.60
+ 3.75
------
  6.35
```

Adding words: in all, total, altogether, combined, both, sum, how much … together.

LET'S TRY ONE

A notebook costs $3.45 and a pen costs $2.80. What is the total cost?

Operation: _____ Estimate: _____

Exact: _____ Answer with units: _____

YOUR TURN

SKILL 4.6 · SOLVE. LABEL YOUR ANSWER.

1 A bag of apples weighs **1.25 lb** and a bag of pears weighs **2.5 lb**. How much do they weigh together?

Answer _____ lb

2 Mia spent **$4.75** on lunch and **$3.60** on a snack. How much did she spend in all?

$ _____

3 A pitcher has **0.6 L** of juice. Sam adds **0.35 L**. How much juice is in the pitcher now?

Answer _____ L

4 A plant was **12.4 cm** tall. It grew **8.75 cm**. How tall is it now?

Answer _____ cm

5 A trail has two parts: **3.2 miles** and **4.85 miles**. How long is the whole trail?

Answer _____ mi

6 A game costs **$15.99**. Shipping is **$6.50**. What is the total?

$ _____

7 Ava biked **2.45 km**, then **1.8 km**, then **0.75 km**. How far did she bike altogether?

Answer _____ km

8 A runner's time was **9.58 seconds**. The next runner was **0.47 second** slower. What was the second runner's time?

Answer _____ seconds

Section 4 Review

Mixed practice from Skills 4.1 to 4.6. No hints: show what you know.

YOUR TURN **REVIEW · PROBLEMS 1–8**

1 **0.34 + 0.52**

Sum _____

2 **6.27 + 1.41**

Sum _____

3 **3.86 + 4.57**

Sum _____

4 **8.4 + 2.75**

Sum _____

5 **10 + 3.06**

Sum _____

6 Estimate by rounding to whole numbers: **5.7 + 3.25**

Estimate _____

7 A juice costs **$2.95** and a muffin costs **$1.75**. What is the total?

$ _____

8 Two bags of flour weigh **0.825 kg** and **0.175 kg**. How much do they weigh together?

Answer _____ kg

Parent Answer Key

Equal values are correct (6.40 = 6.4). Check that every answer has its decimal point lined up. "Try" is the Let's Try One problem.

4.1 Adding with Place Value

Try **0.83** Line up the points. Hundredths: 2 + 1 = 3. Tenths: 5 + 3 = 8. → 0.83.

1 **0.7** Line up the points. Tenths: 2 + 5 = 7. → 0.7.

2 **0.9** Line up the points. Tenths: 6 + 3 = 9. → 0.9.

3 **0.37** Line up the points. Hundredths: 4 + 3 = 7. Tenths: 1 + 2 = 3. → 0.37.

4 **0.77** Line up the points. Hundredths: 1 + 6 = 7. Tenths: 4 + 3 = 7. → 0.77.

5 **4.6** Line up the points. Tenths: 2 + 4 = 6. Ones: 1 + 3 = 4. → 4.6.

6 **6.18** Line up the points. Hundredths: 3 + 5 = 8. Tenths: 1 + 0 = 1. Ones: 2 + 4 = 6. → 6.18.

7 **0.447** Line up the points. Thousandths: 5 + 2 = 7. Hundredths: 0 + 4 = 4. Tenths: 3 + 1 = 4. → 0.447.

8 **4.81** Tenths: 2 + 6 = 8. Hundredths: 1 + 0 = 1. Ones: 3 + 1 = 4. → 4.81.

4.2 Line Up and Add (No Regrouping)

Try **7.57** Line up the points. Hundredths: 1 + 6 = 7. Tenths: 2 + 3 = 5. Ones: 5 + 2 = 7. → 7.57.

1 **3.76** Line up the points. Hundredths: 1 + 5 = 6. Tenths: 3 + 4 = 7. Ones: 2 + 1 = 3. → 3.76.

2 **7.88** Line up the points. Hundredths: 2 + 6 = 8. Tenths: 1 + 7 = 8. Ones: 4 + 3 = 7. → 7.88.

3 **0.77** Line up the points. Hundredths: 3 + 4 = 7. Tenths: 5 + 2 = 7. → 0.77.

4 **9.87** Line up the points. Hundredths: 6 + 1 = 7. Tenths: 0 + 8 = 8. Ones: 7 + 2 = 9. → 9.87.

5 **17.7** Line up the points. Tenths: 3 + 4 = 7. Ones: 2 + 5 = 7. Tens: 1 + 0 = 1. → 17.7.

6 **7.637** Line up the points. Thousandths: 5 + 2 = 7. Hundredths: 0 + 3 = 3. Tenths: 2 + 4 = 6. Ones: 6 + 1 = 7. → 7.637.

7 **18.57** Line up the points. Hundredths: 2 + 5 = 7. Tenths: 4 + 1 = 5. Ones: 5 + 3 = 8. Tens: 1 + 0 = 1. → 18.57.

8 **0.849** Line up the points. Thousandths: 4 + 5 = 9. Hundredths: 1 + 3 = 4. Tenths: 6 + 2 = 8. → 0.849.

4.3 Adding with Regrouping

Try **6.83** Line up the points. Hundredths: 8 + 5 = 13, write 3, carry 1. Tenths: 6 + 1 + 1 (carried) = 8. Ones: 4 + 2 = 6. → 6.83.

1 **3.92** Line up the points. Hundredths: 8 + 4 = 12, write 2, carry 1. Tenths: 5 + 3 + 1 (carried) = 9. Ones: 2 + 1 = 3. → 3.92.

2 **0.87** Line up the points. Hundredths: 8 + 9 = 17, write 7, carry 1. Tenths: 4 + 3 + 1 (carried) = 8. → 0.87.

3 **8.52** Line up the points. Hundredths: 7 + 5 = 12, write 2, carry 1. Tenths: 6 + 8 + 1 (carried) = 15, write 5, carry 1. Ones: 5 + 2 + 1 (carried) = 8. → 8.52.

4 **12.3** Line up the points. Tenths: 6 + 7 = 13, write 3, carry 1. Ones: 9 + 2 + 1 (carried) = 12, write 2, carry 1. Write the carried 1 in the next place on the left. → 12.3.

5 **10.23** Line up the points. Hundredths: 5 + 8 = 13, write 3, carry 1. Tenths: 7 + 4 + 1 (carried) = 12, write 2, carry 1. Ones: 6 + 3 + 1 (carried) = 10, write 0, carry 1. Write the carried 1 in the next place on the left. → 10.23.

6 **8.04** Line up the points. Hundredths: 8 + 6 = 14, write 4, carry 1. Tenths: 9 + 0 + 1 (carried) = 10, write 0, carry 1. Ones: 4 + 3 + 1 (carried) = 8. → 8.04.

7 **20.21** Line up the points. Hundredths: 5 + 6 = 11, write 1, carry 1. Tenths: 8 + 3 + 1 (carried) = 12, write 2, carry 1. Ones: 2 + 7 + 1 (carried) = 10, write 0, carry 1. Tens: 1 + 0 + 1 (carried) = 2. → 20.21.

8 **0.843** Line up the points. Thousandths: 7 + 6 = 13, write 3, carry 1. Hundredths: 8 + 5 + 1 (carried) = 14, write 4, carry 1. Tenths: 5 + 2 + 1 (carried) = 8. → 0.843.

4.4 Different Numbers of Decimal Places

Try **4.77** Line up the points; write 3.50 and 1.27. Hundredths: 0 + 7 = 7. Tenths: 5 + 2 = 7. Ones: 3 + 1 = 4. → 4.77.

1 **3.75** Line up the points; write 2.40 and 1.35. Hundredths: 0 + 5 = 5. Tenths: 4 + 3 = 7. Ones: 2 + 1 = 3. → 3.75.

2 **1.07** Line up the points; write 0.60 and 0.47. Hundredths: 0 + 7 = 7. Tenths: 6 + 4 = 10, write 0, carry 1. Ones: 0 + 0 + 1 (carried) = 1. → 1.07.

3 **7.38** Line up the points; write 5.00 and 2.38. Hundredths: 0 + 8 = 8. Tenths: 0 + 3 = 3. Ones: 5 + 2 = 7. → 7.38.

4 **8.05** Line up the points; write 3.25 and 4.80. Hundredths: 5 + 0 = 5. Tenths: 2 + 8 = 10, write 0, carry 1. Ones: 3 + 4 + 1 (carried) = 8. → 8.05.

5 **0.98** Line up the points; write 0.08 and 0.90. Hundredths: 8 + 0 = 8. Tenths: 0 + 9 = 9. → 0.98.

6 **9.75** Line up the points; write 9.00 and 0.75. Hundredths: 0 + 5 = 5. Tenths: 0 + 7 = 7. Ones: 9 + 0 = 9. → 9.75.

7 **8.725** Line up the points; write 7.125 and 1.600. Thousandths: 5 + 0 = 5. Hundredths: 2 + 0 = 2. Tenths: 1 + 6 = 7. Ones: 7 + 1 = 8. → 8.725.

8 **16.156** Line up the points; write 12.400 and 3.756. Thousandths: 0 + 6 = 6. Hundredths: 0 + 5 = 5. Tenths: 4 + 7 = 11, write 1, carry 1. Ones: 2 + 3 + 1 (carried) = 6. Tens: 1 + 0 = 1. → 16.156.

4.5 Estimate to Check

Try **9** Round each number to the nearest whole: 6.8 ≈ 7, 2.15 ≈ 2. 7 + 2 = 9.

1 **8** Round each number to the nearest whole: 4.8 ≈ 5, 3.1 ≈ 3. 5 + 3 = 8.

2 **8** Round each number to the nearest whole: 2.3 ≈ 2, 5.6 ≈ 6. 2 + 6 = 8.

3 **9** Round each number to the nearest whole: 7.45 ≈ 7, 1.92 ≈ 2. 7 + 2 = 9.

4 **16** Round each number to the nearest whole: 12.6 ≈ 13, 3.4 ≈ 3. 13 + 3 = 16.

5 **9** Round each number to the nearest whole: 0.9 ≈ 1, 8.2 ≈ 8. 1 + 8 = 9.

6 **25** Round each number to the nearest whole: 19.75 ≈ 20, 4.5 ≈ 5. 20 + 5 = 25.

7 **No** Estimate: 4 + 3 = 7. 70.5 is far from 7, so the decimal point is in the wrong place. The exact sum is 7.05.

8 **Yes** Estimate: 6 + 2 = 8. 8.11 is close to 8. (Exact: 5.86 + 2.25 = 8.11.)

4.6 Addition Word Problems

Try **6.25** Total cost means add. Estimate 3 + 3 = 6. 3.45 + 2.80 = 6.25: $6.25.

1. **3.75** Line up the points; write 1.25 and 2.50. Hundredths: 5 + 0 = 5. Tenths: 2 + 5 = 7. Ones: 1 + 2 = 3. → 3.75.
2. **8.35** Line up the points; write 4.75 and 3.60. Hundredths: 5 + 0 = 5. Tenths: 7 + 6 = 13, write 3, carry 1. Ones: 4 + 3 + 1 (carried) = 8. → 8.35.
3. **0.95** Line up the points; write 0.60 and 0.35. Hundredths: 0 + 5 = 5. Tenths: 6 + 3 = 9. → 0.95.
4. **21.15** Line up the points; write 12.40 and 8.75. Hundredths: 0 + 5 = 5. Tenths: 4 + 7 = 11, write 1, carry 1. Ones: 2 + 8 + 1 (carried) = 11, write 1, carry 1. Tens: 1 + 0 + 1 (carried) = 2. → 21.15.
5. **8.05** Line up the points; write 3.20 and 4.85. Hundredths: 0 + 5 = 5. Tenths: 2 + 8 = 10, write 0, carry 1. Ones: 3 + 4 + 1 (carried) = 8. → 8.05.
6. **22.49** Line up the points; write 15.99 and 6.50. Hundredths: 9 + 0 = 9. Tenths: 9 + 5 = 14, write 4, carry 1. Ones: 5 + 6 + 1 (carried) = 12, write 2, carry 1. Tens: 1 + 0 + 1 (carried) = 2. → 22.49.
7. **5.00** Line up the points; write 2.45 and 1.80 and 0.75. Hundredths: 5 + 0 + 5 = 10, write 0, carry 1. Tenths: 4 + 8 + 7 + 1 (carried) = 20, write 0, carry 2. Ones: 2 + 1 + 0 + 2 (carried) = 5. → 5.00 = 5.
8. **10.05** Line up the points. Hundredths: 8 + 7 = 15, write 5, carry 1. Tenths: 5 + 4 + 1 (carried) = 10, write 0, carry 1. Ones: 9 + 0 + 1 (carried) = 10, write 0, carry 1. Write the carried 1 in the next place on the left. → 10.05.

Review Section 4 Review

1. **0.86** Line up the points. Hundredths: 4 + 2 = 6. Tenths: 3 + 5 = 8. → 0.86.
2. **7.68** Line up the points. Hundredths: 7 + 1 = 8. Tenths: 2 + 4 = 6. Ones: 6 + 1 = 7. → 7.68.
3. **8.43** Line up the points. Hundredths: 6 + 7 = 13, write 3, carry 1. Tenths: 8 + 5 + 1 (carried) = 14, write 4, carry 1. Ones: 3 + 4 + 1 (carried) = 8. → 8.43.
4. **11.15** Line up the points; write 8.40 and 2.75. Hundredths: 0 + 5 = 5. Tenths: 4 + 7 = 11, write 1, carry 1. Ones: 8 + 2 + 1 (carried) = 11, write 1, carry 1. Write the carried 1 in the next place on the left. → 11.15.
5. **13.06** Line up the points; write 10.00 and 3.06. Hundredths: 0 + 6 = 6. Tenths: 0 + 0 = 0. Ones: 0 + 3 = 3. Tens: 1 + 0 = 1. → 13.06.
6. **9** Round each number to the nearest whole: 5.7 ≈ 6, 3.25 ≈ 3. 6 + 3 = 9.
7. **4.70** Line up the points. Hundredths: 5 + 5 = 10, write 0, carry 1. Tenths: 9 + 7 + 1 (carried) = 17, write 7, carry 1. Ones: 2 + 1 + 1 (carried) = 4. → 4.70 = 4.7.
8. **1.000** Line up the points. Thousandths: 5 + 5 = 10, write 0, carry 1. Hundredths: 2 + 7 + 1 = 10, write 0, carry 1. Tenths: 8 + 1 + 1 = 10, write 0, carry 1. Ones: 0 + 0 + 1 = 1. → 1.000 = 1 kg.